



Sri Eshwar
College of Engineering
Coimbatore | Tamilnadu
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Affiliated to Anna University, Chennai



Department of Computer Science and Engineering

Academic Year – 2026-27 (Odd Semester)

Project Leap – Java DB Sprint Week

Spring Boot + Database Application Development

Dates: 16.09.2026 – 18.09.2026 & 28.09.2026 - 29.09.2026

Time: 8:40 am – 4:10 pm

Prerequisite Knowledge: Core Java + SQL

Technology: Java, Spring Boot, Maven, REST API, JPA/Hibernate, MySQL, Thymeleaf, Postman



Prerequisites:

Students should have foundational knowledge and comfort with the following:

1. **Java (Core):** Familiarity with OOP concepts such as Classes, Objects, Constructors, Inheritance, Polymorphism, Interfaces, Methods, and basic Exception Handling.
2. **Java Collections (Basics):** Basic understanding of commonly used collections such as ArrayList, HashMap, and how to store and retrieve objects.
3. **SQL (Basics):** Understanding of relational databases, tables, Primary Keys, Foreign Keys, and basic SQL commands such as CREATE, INSERT, SELECT, UPDATE, and DELETE.
4. **Web Concepts (Awareness):** Basic understanding of how a web application works, including the concepts of a browser, client, server, request, and response.
5. **Programming Environment:** Students should be comfortable creating, compiling, running, and debugging basic Java programs using an IDE.

System Setup (Must be completed before Day 1)

All students must have the following software installed and minimally configured on their personal or lab systems:

- **JDK 17+:** Java Development Kit. Java 17 LTS is recommended for a standardized lab environment.
- **Spring Boot:** Latest stable Spring Boot version compatible with the selected JDK.
- **Maven:** Build and dependency management tool for the Spring Boot project.
- **IntelliJ IDEA / VS Code / Eclipse:** IDE for developing and running Spring Boot applications.
- **MySQL 8+:** Relational database server for database integration and CRUD operations.
- **MySQL Workbench:** GUI tool for creating databases, tables, inserting records, and verifying database changes.
- **Postman:** Tool for creating, sending, and testing REST API requests such as GET, POST, PUT, and DELETE.

DAY 1 – Spring Boot Foundation & REST API

Time	Section	Related Tasks	Expected Output
8:40 am – 9:00 am	1. Introduction to the LEAP Project	Identify the project requirements, application workflow and tasks to be completed during the 3-day LEAP	Students identify the application requirements and development roadmap
9:00 am – 9:30 am	2. Why Spring Boot?	Compare the project requirements with Spring Boot features and identify where Spring Boot will be used in the application	Students identify the role of Spring Boot in the project
9:30 am – 10:00 am	3. Spring Boot Application & Maven Basics	Create the project structure, configure pom.xml, add required dependencies and execute the application	Project structure and dependencies configured
10:00 am – 11:15 am	4. Create First Spring Boot Application	Create and run the Spring Boot application; identify project folders and verify successful execution	First Spring Boot application running successfully
11:15 am – 11:35 am	Forenoon Break		
11:35 am – 12:30 pm	5. Spring Boot Annotations & Controller	Create a Controller and implement GET and POST endpoints using @RestController, @GetMapping and @PostMapping.	First controller and GET/POST endpoints created
12:30 pm – 1:15 pm	6. REST API & Request–Response Flow	Develop GET and POST APIs, send requests and verify the returned responses	Basic REST APIs working successfully
1:15 pm – 2:00 pm	Lunch Break		
2:00 pm – 3:00 pm	7. Application Layers – Controller & Service	Create Controller and Service classes and connect the request flow from Controller → Service	Controller–Service request flow implemented
3:00 pm– 4:10 pm	8. API Testing with Postman	Execute GET/POST requests in Postman, verify request parameters and responses, and correct API errors	APIs tested and validated successfully

DAY 2 – Database Integration, JPA & CRUD

Time	Section	Related Tasks	Expected Output
8:40 am – 9:10 am	9. JDBC vs JPA – Why ORM?	Compare a database operation using JDBC and JPA, identify the additional coding required with JDBC, and determine how JPA simplifies database interaction.	Students understand the purpose and practical advantage of JPA/ORM
9:10 am – 10:00 am	10. Database Configuration	Configure MySQL connection in Spring Boot, add the required database and JPA dependencies, configure application.properties, and verify the database connection.	Spring Boot application successfully connected to MySQL
10:00 am – 11:15 am	11. JPA Entity & Repository	Create an Entity class, map Java fields to database columns, configure the primary key, create a JpaRepository, and verify that the corresponding database table is created.	Java Entity successfully mapped to a database table
11:15 am – 11:35 am	Forenoon Break		
11:35 am – 12:30 pm	12. CREATE Operation	Create the Controller, Service and Repository flow; implement a POST API; accept request data and save a new Entity record into MySQL.	New records successfully inserted into the database
12:30 pm – 1:15 pm	13. READ Operation	Implement GET APIs to retrieve all records and a specific record by ID; send requests and verify the returned database records.	Database records successfully retrieved and displayed
1:15 pm – 2:00 pm	Lunch Break		
2:00 pm – 3:00 pm	14. UPDATE & DELETE Operations	Implement PUT and DELETE APIs; update existing records, delete selected records, test both operations, and verify the changes in MySQL.	Records can be successfully updated and deleted
3:00 pm– 4:10 pm	15. Complete CRUD Integration	Use the Postman to verify the corresponding database changes.	Complete CRUD application working

DAY 3 – UI, Validation, Testing & Finalization

Time	Section	Related Tasks	Expected Output
8:40 am – 9:10 am	16. Introduction to Web Interface	Set up the HTML, CSS & Java Script interface, identify the required pages, and connect the UI request flow with the Spring Boot backend.	Students understand the UI flow and backend communication
9:10 am – 10:00 am	17. Create Web Pages & Forms	Create the required web pages and forms; implement fields for adding, viewing, editing and deleting application data.	Basic web interface and forms created
10:00 am – 11:15 am	18. UI–CRUD Integration	Connect the web forms and pages to the Spring Boot CRUD APIs; submit requests from the UI and display the returned data.	CRUD operations accessible through the UI
11:15 am – 11:35 am	Forenoon Break		
11:35 am – 12:30 pm	19. Input Validation	Add validation rules for required fields and invalid input; test valid and invalid form submissions and display appropriate validation messages.	Invalid input handled correctly
12:30 pm – 1:15 pm	20. Exception Handling	Implement handling for common application, API and database errors; test failure scenarios and provide meaningful error responses/messages.	Application errors handled gracefully
1:15 pm – 2:00 pm	Lunch Break		
2:00 pm – 3:00 pm	21. Application Testing & Debugging	Test the complete application workflow, including CRUD operations, form validation and error scenarios.	Major issues identified and fixed
3:00 pm – 4:10 pm	22. Project Finalization & Demonstration	Execute the complete application workflow, verify database operations, perform final testing, prepare the project for demonstration, and demonstrate the completed application.	Complete project tested and ready for demonstration

3-Day Learning Progression

Day	Focus	Students Learn	Final Achievement
Day 1	Spring Boot Foundation	Spring Boot, Maven, Controller, REST, Service, Postman	Build & test REST APIs
Day 2	DB Integration + CRUD	MySQL, JPA, Entity, Repository, CRUD	Build database-backed application
Day 3	UI + Validation + Testing	Thymeleaf, Forms, Validation, Exception Handling, Testing	Complete demonstrable application

Pre-LEAP Orientation session - Maven Project and Introduction to Spring Boot